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A portrait of a sucker using landscape genetics: how colonization and life history undermine the idealized dendritic metapopulation

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Abstract

Dendritic metapopulations have been attributed unique properties by *in silico* studies, including an elevated genetic diversity relative to a panmictic population of equal total size. These predictions have not been rigorously tested in nature, nor has there been full consideration of the interacting effects among contemporary landscape features, colonization history and life history traits of the target species. We tested for the effects of dendritic structure as well as the relative importance of life history, environmental barriers and historical colonization on the neutral genetic structure of a longnose sucker (*Catostomus catostomus*) metapopulation in the Kogaluk watershed of northern Labrador, Canada. Samples were collected from eight lakes, genotyped with 17 microsatellites, and aged using opercula. Lakes varied in differentiation, historical and contemporary connectivity, and life history traits. Isolation by distance was detected only by removing two highly genetically differentiated lakes, suggesting a lack of migration–drift equilibrium and the lingering influence of historical factors on genetic structure. Bayesian analyses supported colonization via the Kogaluk's headwaters. The historical concentration of genetic diversity in headwaters inferred by this result was supported by high historical and contemporary effective sizes of the headwater lake, T-Bone. Alternatively, reduced allelic richness in headwaters confirmed the dendritic structure's influence on gene flow, but this did not translate to an elevated metapopulation effective size. A lack of equilibrium and upstream migration may have dampened the effects of dendritic structure. We suggest that interacting historical and contemporary factors prevent the achievement of the idealized traits of a dendritic metapopulation in nature.

Keywords: colonization history, conservation, dendritic metapopulation, generation time, landscape genetics, migration–drift equilibrium

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Introduction

Habitat fragmentation leads to a decline in gene flow among the resulting subpopulations, causing a loss of overall genetic diversity, and an increased metapopulation extinction risk (Whitlock & Barton 1997; Waples 2010; Palstra & Ruzzante 2011; Gomez-Uchida *et al.* 2013). Metapopulations with a dendritic structure (Fullerton *et al.* 2010; Altermatt 2013) may be an exception to this rule (Morrissey & de Kerckhove 2009). Unlike most metapopulations, the genetically isolated headwater subpopulations in dendritic systems are the source of genetic diversity (Morrissey & de Kerckhove 2009). These headwaters, located at the tips of the dendritic metapopulation (Fagan 2002), exhibit reduced effective population sizes (N_e, Wright 1931) and increased drift (Araki *et al.* 2007; Charlesworth 2009;

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McCracken *et al.* 2013). The unique alleles fixed in each headwater lake collect in downstream confluences leading to their increased genetic diversity and allelic richness (Morrissey & de Kerckhove 2009; Paz-Vinas & Blanchet 2015). The metapopulation effective size of a dendritic system is therefore predicted to be greater than the sum of its subpopulation N_es, in contrast to most other metapopulations (Morrissey & de Kerckhove 2009; Gomez-Uchida *et al.* 2013).

River systems typify this dendritic structure, and fully aquatic organisms confined by the bounds of the river are ideal species for studying dendritic metapopulations (Hughes *et al.* 2009). Within rivers, headwater isolation can be exacerbated by downstream-biased migration (Morrissey & de Kerckhove 2009), reduced carrying capacities in smaller upstream habitats (Carrara *et al.* 2014) and historical upstream colonization from downstream habitats (Paz-Vinas *et al.* 2015). Other environmental factors potentially influencing patterns of diversity include the following: geographic distance (leading to isolation by distance (IBD), Wright 1943), slope and elevation differences (Lowe *et al.* 2006; Caldera & Bolnick 2008; Hughes *et al.* 2009; McCracken *et al.* 2013) and physical barriers (Cote *et al.* 2009; Horreo *et al.* 2011).

Historical colonization can also contribute to the genetic structure of a metapopulation (Castric & Bernatchez 2003; Costello *et al.* 2003; Poissant *et al.* 2005; Vera-Escalona *et al.* 2015). Subsequent to colonization, local environmental factors begin to influence connectivity (Costello *et al.* 2003). With time and an unchanging environment, migration–drift equilibrium will be achieved, leading to IBD (Slatkin 1993; Poissant *et al.* 2005). Because historical patterns will remain in metapopulations that have not yet reached migration–drift equilibrium (Castric & Bernatchez 2003), disentangling the contemporary and historical effects on observed genetic structure is essential in these metapopulations.

A longnose sucker (*Catostomus catostomus* (Forester 1773)) metapopulation within the dendritic Kogaluk River watershed of northern Labrador, Canada (henceforth referred to as the Kogaluk, Fig. 1, Table 1), offers an ideal system to study dendritic metapopulations. Its remoteness minimizes anthropogenic influence (Anderson 1985). A waterfall at the easternmost extent of the drainage closes the system (Anderson 1985) and prevents the confounding effects of immigration.

The longnose sucker is a benthic invertivore found throughout North America and Siberia (McPhail & Lindsey 1970; Ryan 1980; Scott & Crossman 1998). The species is presumed to have colonized northern Labrador from Quebec (Black *et al.* 1986) following the retreat of the Laurentide Ice Sheet which occurred approximately 9000 years ago (Bryson *et al.* 1969; Short & Nichols 1977). It has been hypothesized that the Kogaluk was colonized from the west via overflow events from the palaeolake Naskaupi which was present to the west of the Kogaluk between 8400 and 7500 years BP (Jansson & Kleman 2004). Alternatively, this species may have invaded from the coast given its tolerance for brackish water (Black *et al.* 1986). Presently, longnose suckers inhabit a number of lakes connected by shallow streams within the Kogaluk. In addition to the Kogaluk's dendritic structure, waterfalls, elevation differences and geographic distance among the lakes may influence this species' genetic structure. Upstream dispersal of this species during spring spawning events (Harris 1962; McPhail & Lindsey 1970; Ryan 1980; Walton 1980; Scott & Crossman 1998) could erode headwater isolation, countering the effects of the dendritic structure (Morrissey & de Kerckhove 2009).

Although the proposed properties of dendritic metapopulations are potentially paradigm-shifting for the field of landscape genetics (Manel *et al.* 2003; Storfer *et al.* 2007), they are largely based on modelling studies with little testing of these properties in natural systems (Campbell Grant *et al.* 2007; Labonne *et al.* 2008; Campbell Grant 2011; Perkin & Gido 2012; Altermatt 2013). Similarly, there has been little comprehensive study of how history interacts with environmental and biological factors (Costello *et al.* 2003; Orsini *et al.* 2013). In the present study, we employed microsatellite and life history data to (i) examine the extent to which the theoretical properties of dendritic metapopulations are actually observed in natural systems and (ii) identify the relative effects of life history, environmental barriers, dendritic structure and colonization history on a metapopulation's genetic structure.

Methods

Study site and sampling

The Kogaluk comprises hierarchically arranged postglacial lakes that drain into the fjord lake, Cabot lake, which in turn drains into the Atlantic Ocean (Anderson 1985). Several waterfalls (>5 m high) are putative barriers to upstream fish migration (Anderson 1985). These waterfalls isolate Cabot lake from its upstream tributaries and the Atlantic Ocean (Anderson 1985).

Longnose sucker samples were collected between 2006 and 2014 from eight lakes within the Kogaluk. To sample a variety of age classes, lakes were sampled at two or more locations using variably sized standardized nylon monofilament gillnets (1.27 cm to 8.89 cm diagonal) and/or electrofishing. Fish were weighed,

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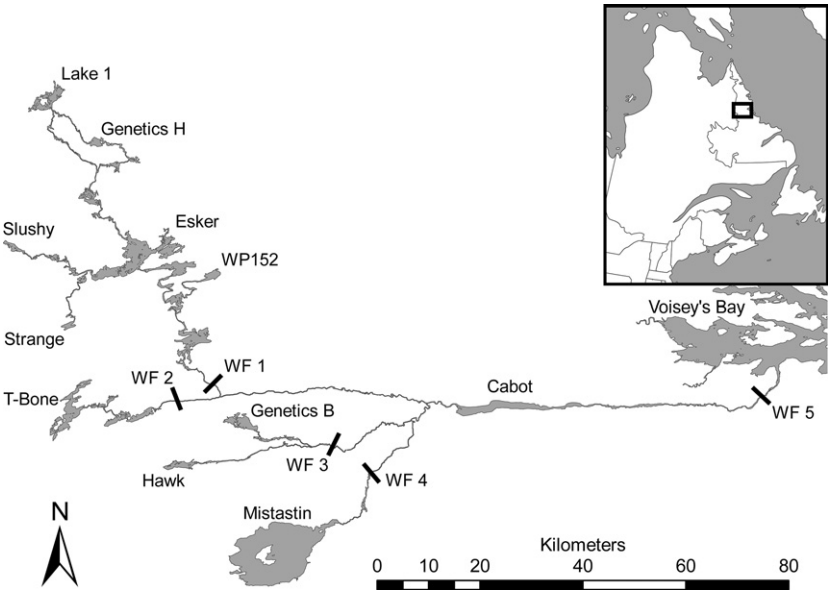

Fig. 1 The Kogaluk drainage in northern Labrador. The map inset demonstrates the location of the system within Labrador. The Kogaluk drains into the Atlantic Ocean by Voisey's Bay at the easternmost extent of the system. Five waterfalls (WF1-WF5) are present within the system at the following heights: WF1 is 15 m high, WF2 is 12 m, WF3 is 5.4 m, WF4 is 5.4 m, and WF5 is 9 m (Anderson 1985). Map created in ArcMap 10 (ESRI, Redlands, CA, USA).

measured for fork length (FL) in centimetres, and assessed for sex and maturity. Pectoral fin clips were clipped and either immediately stored in 95% ethanol or stored dry (n = 1297). Opercula were extracted, cleaned in warm water and stored dry. Opercula for young-of-the-year fish were not collected due to their fragility and removal difficulty. Fish < 70 mm were assumed to be young-of-the-year. Opercula were aged following Perry & Casselman (2012) by observing them concave side up under a dissecting microscope. Perry & Casselman (2012) suggest an age correction to compensate for the number of annuli covered by the dense bone growth region. We conducted analyses with both corrected and uncorrected ages, and results were similar. For simplicity, we present analyses with the uncorrected ages only. Analyses conducted with corrected ages are presented in Appendix S1, Supporting Information.

Life history analyses

Age at 50% maturity (α) was estimated using a binomial logistic regression (Harry *et al.* 2013) with age as the independent variable and maturity (0 as immature, 1 as mature) as the dependent variable (n = 1072) in R (R Core Team, 2013). Adult lifespan (AL) was estimated as $AL = \omega - \alpha + 1$, where ω is the maximum age observed from the sampled lake (Waples *et al.* 2014).

Age distributions of gillnet-caught samples were compared between years within lakes using Kolmogorov–Smirnov tests with Bonferonni-corrected α-values. Samples were pooled for each lake when yearly age distributions were not significantly different. The Robson–Chapman annual survivorship estimate ($\hat{S}$) (Chapman & Robson 1960; Robson & Chapman 1961) was calculated for each lake using ages of gillnet-caught samples (n = 1050). $\hat{S}$ was used to determine l_x (probability of survival to age x) for each age class (x), assuming $l_0 = 1$, $l_1 = \hat{S}^1 * l_0, \ldots l_\omega = \hat{S}^\omega * l_0$.

The ages of mature females were $\log_e$-transformed and correlated with FL using a linear model (n = 204). In order to relate age with fecundity, FL was equated to total length in the following equation relating fecundity (f is the number of eggs produced) with total length in centimetres (TL) (Childress *et al.* 2015) for longnose suckers in the Great Lakes.

$$f = 0.016 * (\mathrm{TL})^{3.799} \qquad \text{(eqn 1)}$$

Our assumption that this equation is applicable to longnose suckers in the Kogaluk is supported by Blanck & Lamouroux (2007) who found that fecundity was not significantly correlated with latitude in 20 species of European freshwater fish.

Substituting FL for TL in eqn (1) was justified as FL and TL are often linearly related (Carlander & Smith 1945), and therefore, the resulting f would be biased by a constant factor for each age class, which would cancel out with the calculation of generation time.

Fecundity was estimated for each age class from α (rounded down to the nearest age) to ω and divided by 2 to approximate m_x (the number of offspring produced by an individual of age x) to account for the fact that only half of the population is female.

Generation time in years (T) was calculated according to Birch (1948) as:

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Table 1 Environmental, life history and genetic characteristics of eight lakes within the Kogaluk River (Labrador, Canada). MAS is metres above sea level, N/A is not available, ω is maximum age observed in each lake, AL is adult lifespan for each lake, $\hat{S}$ is the Robson–Chapman survivorship estimate, CI is confidence interval, T is generation time, N is the number of genetic samples, A_R is allelic richness, H_o is observed heterozygosity, H_e is expected heterozygosity

Lake name	Latitude, longitude	MAS	Area (km^2)	Depth (m)	ω (years)	AL (years)	$\hat{S}$ (95% CI)	T (years)	N	A_R	H_o	H_e	$\hat{N}_{e(LDNe)}$ with migrants (95% CI)	Migrants removed	$\hat{N}_{e(LDNe)}$ without migrants (95% CI)
Lake 1	56°40′ 31.7″N, 64°00′ 07.5″W	525	11.3	3.9	32	22.3	0.704 (0.613–0.794)	13.3	59	8.28	0.66	0.65	557.6 (202.5 – ∞)	4	704.3 (215.4 – ∞)
Genetics H	56°36′ 13.7″N, 63°52′ 09.1″W	512	2.81	6.5	25	15.3	0.706 (0.654–0.759)	13.2	201	9.45	0.63	0.64	168.1 (134.9–216.8)	2	168.0 (133.6 – 219.5)
Slushy	56°24′ 56.2″N, 64°06′ 08.1″W	464	2.99	15.3	26	16.3	0.602 (0.518–0.685)	12.0	103	9.92	0.65	0.64	313.7 (209.2–590.2)	5	256.8 (178.4 – 436.1)
Strange	56°17′ 24.8″N, 63°56′ 53.4″W	487	2.09	N/A	22	12.3	0.716 (0.666–0.766)	13.1	122	8.48	0.61	0.61	821.0 (382.8 – ∞)	5	1301.1 (451.7 – ∞)
Esker	56°24′ 53.4″N, 63°40′ 15.1″W	431	53.98*	N/A	51	41.3	0.910 (0.894–0.927)	24.2	138	11.19	0.65	0.65	2740.4 (1017.8 – ∞)†	0	2802.6 (1060.7 – ∞)†
WP152	56°22′ 08.7″N, 63°29′ 30.5″W	445	53.98*	16.1	36	26.3	0.795 (0.740–0.850)	15.6	74	11.01	0.68	0.67	2740.4 (1017.8 – ∞)†	3	2802.6 (1060.7 – ∞)†
T-Bone	56°09′ 09.7″N, 63°56′ 21.2″W	468	19.76	N/A	16	6.3	0.692 (0.609–0.775)	12.0	115	9.62	0.63	0.64	−39298.7 (820.3 – ∞)	1	11014.7 (762.5 – ∞)
Cabot	56°08′ 27.9″N, 62°37′ 52.4″W	60	25.39	N/A	21	11.3	0.623 (0.553–0.693)	12.1	57	10.52	0.65	0.65	1196.6 (302.1 – ∞)	1	4287.2 (332.3 – ∞)

*Habitat area estimate for Esker and WP152 combined.
†$\hat{N}_{e(LDNe)}$ estimated for samples from both Esker and WP152.

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$$T = \frac{\sum x l_x m_x}{\sum l_x m_x} \quad \text{(eqn 2)}$$

DNA extraction, amplification and genotyping

Fin clips were digested at 55 °C for approximately 8 h using Proteinase K (Bio Basic Inc., Markham, ON, Canada). DNA was then extracted using a Multiprobe II plus liquid handling system (PerkinElmer, Waltham, MA, USA) using a glassmilk protocol modified from Elphinstone *et al.* (2003).

Twenty microsatellite loci (described in Tranah *et al.* 2001; Lippe *et al.* 2004; Cardall *et al.* 2007; McCracken *et al.* 2014) with demonstrated polymorphisms that could be accurately and consistently scored were selected for amplification (see Table S1 for list of loci, Supporting Information).

Samples were amplified using PCR (see Table S2 for reaction reagents and Table S3 for thermocycler conditions, Supporting Information), visualized using Li-COR 4200/4300 DNA Analyzers (Li-COR Biosciences, Lincoln, NE, USA) and genotyped using SAGA AUTOMATED MICROSATELLITE SOFTWARE 3.3 (Li-COR Biosciences, Lincoln, NE, USA).

Scoring errors and null allele presence was assessed using MICROCHECKER 2.2.3 (van Oosterhout *et al.* 2004). Linkage disequilibrium and departures from Hardy–Weinberg equilibrium (HWE) were assessed using ARLEQUIN 3.5.1.3 (Excoffier & Lischer 2010) (see Appendix S3, Supporting Information for details).

Genetic analyses

FSTAT (Goudet 2001) was used to estimate allele frequencies and allelic richness (A_R) per lake, and Arlequin was used to estimate each lake's observed (H_o) and expected heterozygosity (H_e). We correlated A_R, H_o and H_e with waterway distance to the most downstream lake (Cabot). $\hat{F}_{ST}$s were estimated in MSA 4.05 using 100 000 MCMC permutations (Dieringer & Schlötterer 2003). A principal coordinates analysis (PCoA) using linearized pairwise $\hat{F}_{ST}$s ($\hat{F}_{ST}/(1-\hat{F}_{ST})$) (Rousset 1997) was conducted using GENALEX 6.501 (Peakall & Smouse 2006).

The genetic distances of each stream section were assessed using STREAMTREE analysis of pairwise $\hat{F}_{ST}$s (Kalinowski *et al.* 2008). Because STREAMTREE allows only one pathway between populations, and Lake 1 and Genetics H are connected by two different routes, a separate analysis was conducted for each possible connection between these two lakes. Population structure was assessed using hierarchical analyses conducted in STRUCTURE 2.3.4 (Pritchard *et al.* 2000; Hubisz *et al.* 2009, see Appendix S3, Supporting Information for details.

Contemporary gene flow between lakes was estimated using BAYESASS version 3.0 (which estimates migration over the last several generations, Wilson & Rannala (2003)) (see Appendix S3, Supporting Information for details).

Estimating causes of genetic differentiation

A series of Mantel tests each using 9999 iterations were conducted in GENALEX to test for relations between linearized $\hat{F}_{ST}$s ($\hat{F}_{ST}/(1-\hat{F}_{ST})$) (Rousset 1997) and lake pairwise waterway distance, differences in elevation, slope (calculated according to Stelkens *et al.* 2012) and the number of waterfalls between lakes. Waterway distances were calculated using Google Earth (Google, Mountain View, CA, USA) and ARCMAP 10 (ESRI, Redlands, CA, USA). Decomposed pairwise regression was used after each test to identify and remove potential outlier lakes, which could be masking the effects of the tested landscape variable (Koizumi *et al.* 2006; McCracken *et al.* 2013). Partial Mantel tests of significant variables were conducted in R.

Given the recent controversy of the use of Mantel tests in landscape genetics studies (Diniz-Filho *et al.* 2013; Legendre *et al.* 2015), distance-based redundancy analysis (dbRDA) was also used to determine environmental factors affecting genetic distance between lakes as suggested by Kierepka & Latch (2015). Lake elevation, waterfall distance and waterway distance were used as explanatory variables and compared both individually and together with the principal coordinates of the pairwise $\hat{F}_{ST}$s using dbRDA. Waterway distance variables were calculated from the raw pairwise waterway distance matrix by selecting principal coordinate analysis-derived eigenvectors using the B4-broken stick criterion described by King & Jackson (1999). Significant explanatory variables in those significant dbRDA with multiple explanatory variables were determined using the forward.sel() function from the PACKFOR package (Dray *et al.* 2007; Blanchet *et al.* 2008).

To test whether drift was responsible for lake genetic differentiation, mean pairwise A_R was correlated with pairwise $\hat{F}_{ST}$s. A negative correlation would indicate that populations with low A_R were genetically distinct due to drift (Raeymaekers *et al.* 2008). Additionally, mean pairwise A_R was correlated with standardized residual linearized pairwise $\hat{F}_{ST}$s after the regression of linearized pairwise $\hat{F}_{ST}$s with geographic distance to determine whether drift could explain underlying variation in any detected pattern of IBD.

$\hat{N}_e$ and $\hat{N}_b$ estimation

$\hat{N}_e$ was calculated for each lake using LDNe (Waples & Do 2008) with and without migrants identified in

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GENECLASS2 (Piry *et al.* 2004) to meet the assumption of closed subpopulations (Waples & Do 2008) (see Appendix S3, Supporting Information for GENECLASS2 analysis details). For all LDNe analyses, a P_{crit} (minimum allele frequency) of 0.02 was used as advised by Waples & Do (2010) as all lakes had a sample size >25.

The metapopulation N_e was estimated using the method described by Tufto & Hindar (2003) (*meta*-$\hat{N}_{e(T+H)}$). This technique has fewer assumptions than others and is more precise when metapopulation N_e is relatively large (Gomez-Uchida *et al.* 2013). *Meta*-$\hat{N}_{e(T+H)}$ were estimated using LDNe-derived $\hat{N}_e$ ($\hat{N}_{e(LDNe)}$) generated with and without migrants. Migration rates for each *meta*-$\hat{N}_{e(T+H)}$ were estimated in BAYESASS using the same parameters as described above.

The sum of the lake $\hat{N}_{e(LDNe)}$ was compared with *meta*-$\hat{N}_{e(T+H)}$. A larger *meta*-$\hat{N}_{e(T+H)}$ would indicate dendritic structure conferred an elevated genetic diversity in comparison with an equivalent panmictic population as predicted by Morrissey & de Kerckhove (2009).

Correlations between lake $\hat{N}_e$ and A_R with each of elevation and lake area were conducted in R. A negative correlation between $\hat{N}_e$ or A_R and elevation would indicate downstream-biased dispersal (Castric *et al.* 2001; McCracken *et al.* 2013). A positive correlation between $\hat{N}_e$ or A_R and lake area would indicate habitat area and genetic drift dictates genetic diversity (McCracken *et al.* 2013).

The effective number of breeders ($\hat{N}_b$) was estimated for each lake using LDNe. A combination of two or three age cohorts was used to estimate $\hat{N}_b$ when a single cohort did not yield a positive value, as suggested by Waples *et al.* (2014).

To correct for age structure bias, raw $\hat{N}_b$ were adjusted based on the life history traits, age at first maturation (α), and adult lifespan (AL) according to Waples *et al.* (2014):

$$\hat{N}_{b(adj2)} = \frac{\text{raw}\hat{N}_b}{1.103 - 0.245 * \log\left(\frac{AL}{\alpha}\right)} \qquad \text{(eqn 3)}$$

These adjusted estimates of effective number of breeders ($\hat{N}_{b(adj2)}$) were subsequently used to estimate adjusted effective population size estimates ($\hat{N}_{e(adj2)}$):

$$\hat{N}_{e(adj2)} = \frac{\hat{N}_{b(adj2)}}{0.485 + 0.758 * \log\left(\frac{AL}{\alpha}\right)} \qquad \text{(eqn 4)}$$

Historical migration and N_e estimation

Historical mutation-scaled migration rates (M) and effective sizes (Θ) were estimated using coalescent Bayesian modelling in 3.6 MIGRATE-N (Beerli 2006).

Estimations of M (m/μ) and Θ ($xN_e\mu$, where x is the ploidy multiplier, $x = 4$ for nuclear data) were made using a Brownian motion model. We used one long chain with three replicates, a burn-in of 5.0×10^4, and an increment step of 50 for a total of 7.5×10^6 visited parameter values. We used a static heating scheme consisting of four chains with temperatures: 1.0×10^7, 3.00, 1.50, 1.00. M and Θ each followed exponential prior distributions. Θ ranged between 0 and 100, with mean = 10. M ranged between 0 and 10 000, with mean = 100.

Two migration models were compared in MIGRATE-N, one allowing migration between all lakes and the other allowing migration only between contemporarily connected lakes and those lakes potentially connected by Naskaupi (i.e. the westernmost headwaters: T-Bone, Strange, Slushy) in the light of DIYABC analysis evidence for colonization from the west. In the second model, migration rates were allowed to vary and bidirectional migration was assumed between connected lakes with the exception of Cabot from which no upstream migration was allowed. The log marginal likelihood using Bezier scores was used to determine the best-fitting model (Beerli & Palczewski 2010).

Historical colonization assessment

DIYABC v2.0 (Cornuet *et al.* 2014) was used to assess the relative likelihood of three potential colonization routes (Fig. 2). The first depicted invasion, that is colonization from the south and east, is based on the prediction that longnose suckers colonized northern Labrador via the coast (Black *et al.* 1986). In this scenario, the most downstream lake Cabot was colonized first, followed by the downstream lakes Esker-WP152 at time t_e from which all other headwaters were colonized at later time intervals (t_a, t_b, t_c, t_d). The southwestern-most headwater T-Bone was independently colonized from Cabot at some other time t_f ($t_f \geq t_e$, $t_f \geq t_d \geq t_c \geq t_b \geq t_a$, and $t_e > t_d \geq t_c \geq t_b \geq t_a$).

The second colonization route was based on the STRUCTURE results, where Lake 1, Genetics H, Slushy, Strange, Esker-WP152 and T-Bone were initially colonized at the same time t_h. Cabot was subsequently colonized from Esker-WP152 at time t_g in accordance with the location priors STRUCTURE results ($t_h > t_g$).

A third scenario depicted colonization from the south and west via the palaeolake Naskaupi. The most westerly headwaters (T-Bone, Strange and Slushy) were first colonized at time t_m. These lakes were either covered by Naskaupi or immediately adjacent to this lake between 8400 and 7000 years BP (Jansson & Kleman 2004). Migrants from Strange and Slushy invaded the confluence Esker-WP152 at time t_l from which the more northern headwater lakes Genetics H and Lake 1 were subsequently invaded (at times t_k and t_j, respectively).

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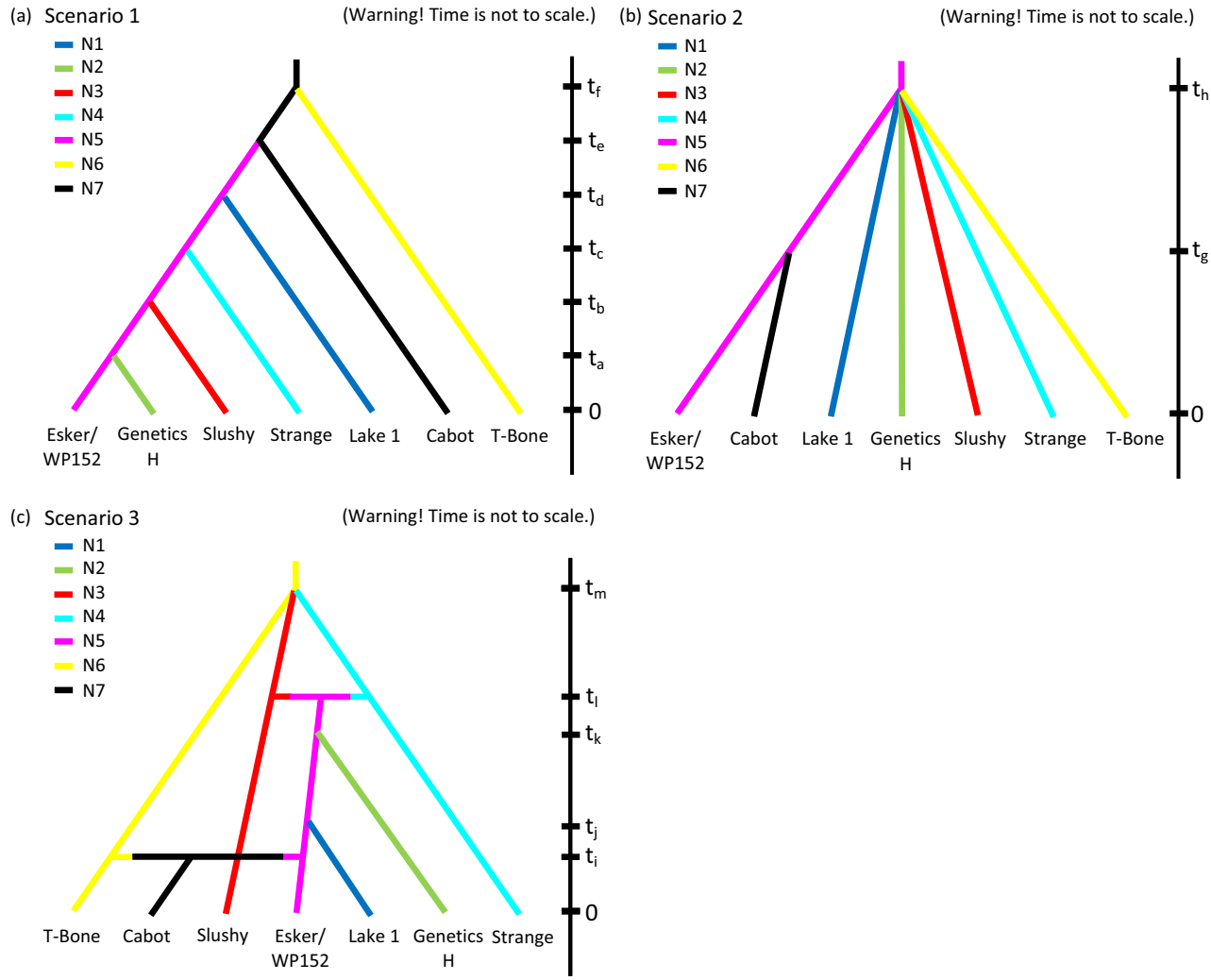

Fig. 2 Three scenarios outlining colonization by longnose suckers (*Catostomus catostomus*) into the Kogaluk River assessed using DIYABC. Time in generations is t. (a) Scenario 1: Cabot was colonized first followed by the colonization of Esker-WP152 from which all other lakes in the northern arm of the Kogaluk were colonized. T-Bone was colonized from Cabot at some independent time ($t_f \geq t_e$, $t_f \geq t_d \geq t_c \geq t_b \geq t_a$, and $t_e > t_d \geq t_c \geq t_b \geq t_a$). (b) Scenario 2: based on the STRUCTURE results where all lakes were colonized at the same time with the exception of Cabot which was colonized from Esker-WP152 at some more recent time ($t_h > t_g$). (c) Scenario 3: colonization occurred first in the westernmost lakes (T-Bone, Slushy and Strange) from lake Naskaupi. Esker-WP152 was colonized from an admixture event of migrants from Slushy and Strange. Genetics H and Lake 1 were colonized from Esker-WP152 subsequently. Cabot was colonized as an admixture event of migrants from T-Bone and Esker-WP152 ($t_m > t_l > t_k \geq t_j \geq t_i$).

The most downstream lake Cabot was invaded from the lakes immediately upstream from it (Esker-WP152 and T-Bone) at time t_i (Accordingly, $t_m > t_l > t_k \geq t_j \geq t_i$).

Colonization was assumed to occur ca. 9000 years BP after the Laurentide Ice Sheet retreated from the Kogaluk (Bryson *et al.* 1969; Short & Nichols 1977). Given that generation time, T, for most lakes varied between 12 and 13.5 years (See Results, Table 1) all time points were set prior distributions between 10 and 750 generations.

In total, 3×10^6 simulations were run for all three models. A stepwise mutation model was assumed and mean mutation rate across loci was assigned a uniform distribution and was allowed to range between 1×10^{-4} and 1×10^{-3}, while individual locus mutation rate was allowed to range between 1×10^{-5} and 1×10^{-2}. N_es were allowed to be variable among lakes, to range uniformly between 1×10^1 and 5×10^4, and were assumed to be equal through time. The one-sample summary statistics employed for generation of simulated data sets included the following: mean number of alleles, mean genic diversity and mean size variance. Two-sample summary statistics included mean genic diversity, mean size variance and $(d\mu)^2$ distance.

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A principal components analysis was used to pre-evaluate the similarity between scenario-generated data sets and the observed data set. The posterior probabilities of all three scenarios were assessed with the logistic regression method using a subset of 1% of the closest simulated data. Linear regression was used to determine the logit-transformed posterior parameters' distributions, using 1% of the closest simulated data sets. Bias and precision were estimated for each scenario using 500 pseudo-observed test data sets simulated using the original parameters from the 1% subset of the closest simulated data sets. Type I and type II error rates were generated for each scenario using confidence estimates derived from 500 pseudo-observed test data sets simulated using the original parameters. Model checking was completed for each scenario using two summary statistics not used in the initial data set generation as suggested by Cornuet *et al.* (2010): the two-sample mean number of alleles and $\hat{F}_{ST}$.

Results

Life history

Opercula were used to age $n = 1158$ individuals, and $n = 195$ were identified as young-of-the-year based on fork length (<70 mm) for a total of $n = 1353$ aged samples. Ages ranged from young-of-the-year to 51, α was 10.7 (95% CI = (10.3, 11.0)) (Fig. S1, Supporting Information), AL ranged from 6.3 in T-Bone to 41.3 in Esker.

The natural logarithm transformation for ages of mature females was significantly correlated with FL ($P < 0.001$, R^2 value of 0.62):

$$\mathrm{FL} = 13.94 * \log_e(\mathrm{Age}) + 0.98) \qquad \text{(eqn 5)}$$

Substitution of FL in eqn (5) with TL in eqn (1) resulted in the following equation used to generate f for each age class:

$$f = 0.016 * [13.94 * (\log_e(\mathrm{Age}) + 0.98)^{3.799}] \qquad \text{(eqn 6)}$$

Age distributions did not differ between sampling years in any of the lakes (Kolmogorov–Smirnov tests all $P > 0.05$, Table S4, Supporting Information); therefore, annual samples within lakes were pooled. Resulting annual survivorship rate estimates, $\hat{S}$, varied between 0.602 in Slushy and 0.910 in Esker, generation time, T, varied from 12.0 in T-Bone and Slushy to 24.2 in Esker.

Genetic quality control

A total of $n = 869$ individuals were successfully amplified for at least 13 of 20 markers. Two loci (DLU4259 and MOHU268) had low amplification success (>10% missing over all individuals), and a third locus (DLU4183) exhibited null alleles in six lakes. These three markers were excluded from analysis. Remaining loci demonstrated no consistent null alleles or departures from HWE and were retained for further analysis (Appendix S3, Supporting Information).

Genetic characteristics

Allelic richness, A_R, correlated negatively with distance to the most downstream lake (Cabot) in the northern six lakes ($R^2 = 0.74$, $P \leq 0.028$) but not in all lakes ($R^2 = 0.33$, $P \geq 0.133$). Neither H_o nor H_e correlated with distance to the most downstream lake in the full system or the northern six lakes (all $P > 0.05$, see Fig. S2, Supporting Information).

Genetic population structure

The PCoA of linearized $\hat{F}_{ST}$s indicated the genetic distinctiveness of Strange and to a lesser extent Lake 1 from the other lakes (Fig. 3a). T-Bone was also separated from the northern lakes and closest to Cabot, which is the lake it drains into. The downstream and immediately adjacent Esker and WP152 were genetically similar to each other.

Genetic distances of stream sections calculated in STREAMTREE were highest between two headwater lakes (Strange and Lake 1) and the confluence Esker; the lowest were between Esker and WP152 and upstream of the most downstream lake Cabot (Fig. 3b, c). The model where the most northern two lakes (Lake 1 and Genetics H) were connected at a point downstream from both lakes had more support ($R^2 = 0.948$) than the model in which Lake 1 drained directly into Genetics H ($R^2 = 0.879$). Distances to Lake 1 were therefore calculated assuming the former model for the IBD Mantel tests.

The Evanno *et al.* (2005) method suggested $K = 6$ for the initial STRUCTURE analysis (Fig. 3d). All headwater lakes were genetically distinct, but the confluences (Esker, WP152 and Cabot) were indistinguishable from each other. Further genetic differentiation among these three lakes was found with the inclusion of location information in the Bayesian model. This was justified because the great distance and intermediate waterfall separating Esker and WP152 from Cabot suggest a lack of panmixia. With the use of location priors, Cabot was genetically distinguishable from both Esker and WP152 ($K = 2$) which were in turn genetically indistinguishable.

Causes of genetic differentiation

A Mantel test of linearized pairwise $\hat{F}_{ST}$s and waterway distances was initially nonsignificant ($R^2 = 0.12$, $P \geq 0.138$). However, a significant correlation was evident after the removal of Strange ($R^2 = 0.38$, $P \leq 0.011$)

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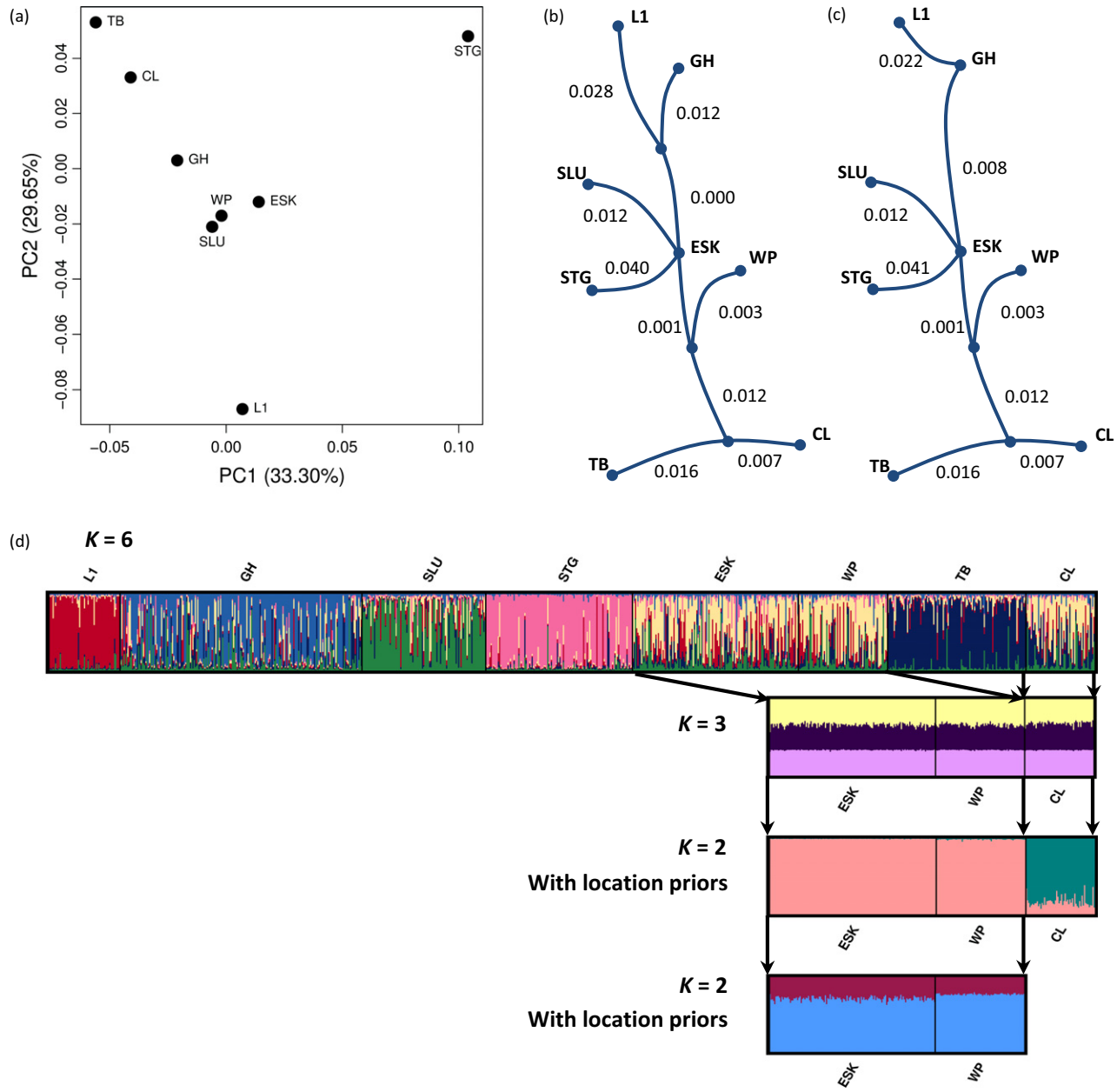

Fig. 3 Three analyses of longnose sucker genetic structure in eight lakes within the Kogaluk River: Lake 1 (L1), Genetics H (GH), Slushy (SLU), Strange (STG), Esker (ESK), WP152 (WP), T-Bone (TB) and Cabot (CL). (a) Principal coordinates analysis based on linearized pairwise $\hat{F}_{ST}$s. Genetic distances associated with stream sections calculated using STREAMTREE analysis of $\hat{F}_{ST}$s assuming. (b) Lake 1 and Genetics H are connected at a common downstream fork and (c) Lake 1 drains directly into Genetics H. (d) Hierarchical STRUCTURE plot. *Esker and WP152 were combined for LDNe estimate of effective size.

and subsequently Lake 1 ($R^2 = 0.77$, $P \leq 0.003$), which were identified as outlier lakes by decomposed pairwise regression analysis (Fig. 4a). The removal of both lakes resulted in the lowest AIC_c (see Table S5, Supporting Information).

Neither elevation nor slope correlated with genetic distance even after the removal of outlier lakes (all P values > 0.05). The Mantel test correlating the number of waterfalls with genetic distance was significant when Strange and Lake 1 were removed. This model also had the lowest AIC_c ($R^2 = 0.70$, $P \leq 0.027$).

Given the collinearity between the number of intermediate waterfalls and waterway distances when Strange and Lake 1 were removed ($R^2 = 0.67$, $P \leq 0.033$), two partial Mantel tests were conducted to determine which variable correlated best with genetic distance. Waterway and genetic distances were correlated after controlling for the number of waterfalls

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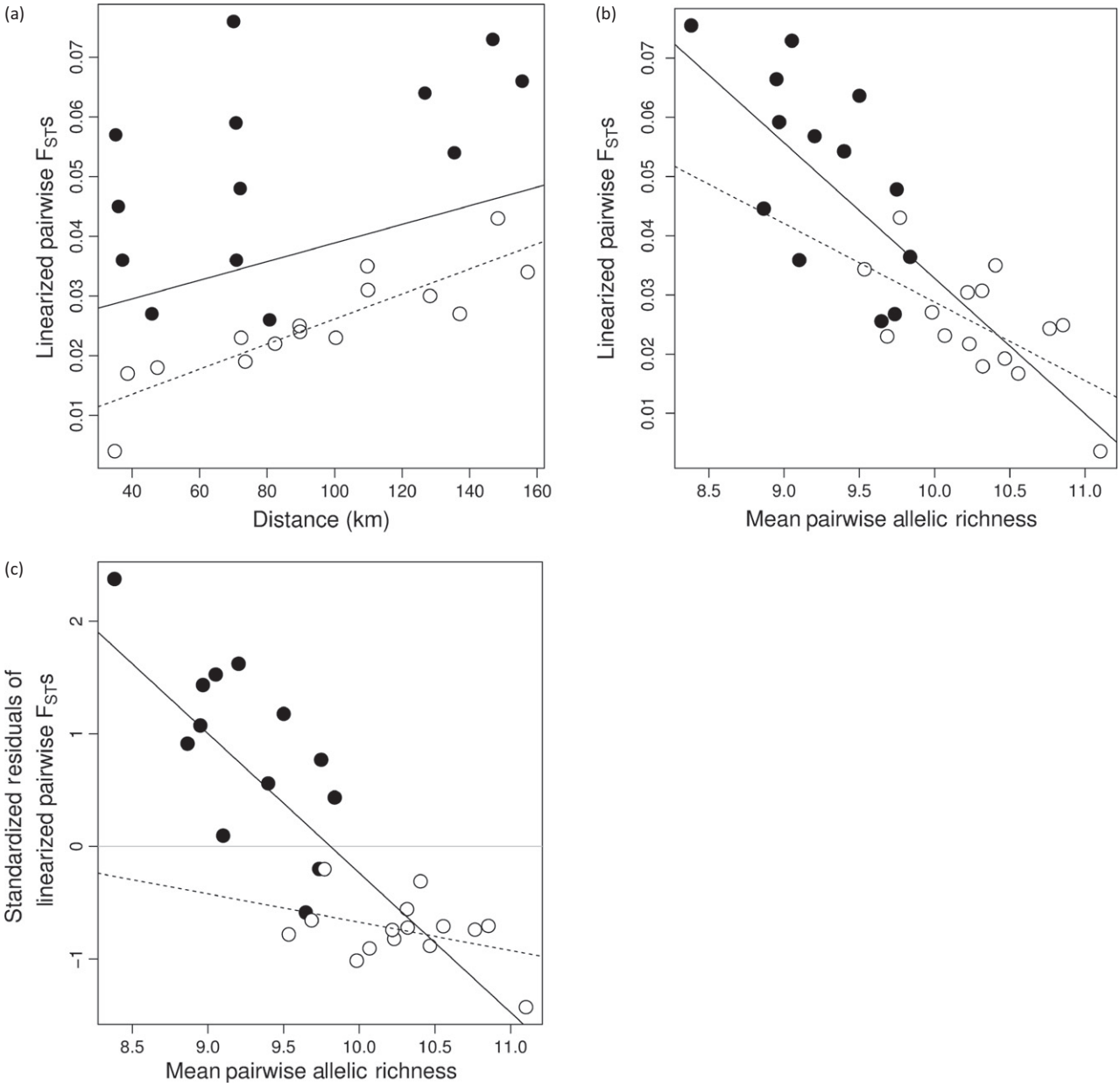

Fig. 4 Correlation of linearized pairwise $\hat{F}_{ST}$s with (a) waterway distance (km) and (b) mean pairwise allelic richness. (c) Correlation of mean pairwise allelic richness with linearized pairwise $\hat{F}_{ST}$ residuals (after regression of $\hat{F}_{ST}$ with geographic distance). Black points were associated with the outlier lakes Strange and Lake 1 identified using decomposed pairwise regression of IBD Mantel tests. White points were associated with nonoutlier lakes. The black solid line indicates the correlation including all lakes. The dashed line indicates the correlation including only nonoutlier lakes (i.e. excluding Strange and Lake 1).

($R^2 = 0.60$, $P \leq 0.032$). However, the number of waterfalls did not correlate with genetic distance after controlling for waterway distance ($R^2 = 0.432$, $P \leq 0.067$), suggesting the greater influence of waterway distances over that of the presence of waterfalls.

The dbRDA provided similar results to the Mantel tests. Only the first eigenvector of the principal coordinates of the pairwise waterway distance matrix was retained after applying the B4-broken stick criterion. When waterfall distance, elevation and the first waterway distance principal eigenvector were included as explanatory variables, the dbRDA was insignificant ($P \geq 0.535$, Table S7, Supporting Information). When considered individually, only the first waterway distance principal eigenvector significantly correlated with the genetic principal coordinates ($P \leq 0.013$). Effects of elevation and waterfall distance were both insignificant ($P \geq 0.42$, 0.09, respectively).

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Mean pairwise A_R was negatively correlated with both linearized pairwise $\hat{F}_{ST}$s and its standardized residuals after correlation with geographic distances ($R^2 = 0.69$, $P \leq 0.001$; $R^2 = 0.69$, $P \leq 0.001$ respectively, Fig. 4b, c). The correlation between mean pairwise A_R and linearized $\hat{F}_{ST}$s remained significant with the removal of Strange and Lake 1 ($R^2 = 0.39$, $P \leq 0.013$). This suggests drift greatly affects this metapopulation and causes lake genetic divergence by reducing their A_R. However, the correlation between mean pairwise A_R and standardized residuals of linearized $\hat{F}_{ST}$s became insignificant when Strange and Lake 1 were removed ($R^2 = 0.02$, $P \geq 0.637$). This indicates that Strange and Lake 1 experience more drift (and loss of A_R) than the other lakes, once pairwise distances are accounted for.

Contemporary and historical migration and genetic diversity

Significant contemporary gene flow calculated using BAYESASS ranged from 3% per generation from the headwater Genetics H to the confluence Esker and from Esker to the headwater, Strange, to 30% from Esker to the nearby downstream lake WP152 (Fig. 5). Most movement in the system was emigration from Esker to both headwater and downstream lakes (Slushy (8%), Strange (3%), Genetics H (5%), WP152 (30%), Cabot (24%)).

Twenty-three individuals were identified as migrants using GENECLASS2. $\hat{N}_e$ were calculated for each lake with and without migrants. Esker and WP152 samples were combined given their genetic similarity. Migrants sampled in WP152 but with a putative origin of Esker, and vice versa, were treated as residents of this combined population and not removed when excluding migrants.

Positive $\hat{N}_e$s were estimated for all lakes except for the most western headwater, T-Bone, when migrants were included; however, upper confidence intervals for 14 of 16 $\hat{N}_e$ values were infinite (Table 1). Furthermore, when excluding migrants, T-Bone's $\hat{N}_e$ was near the upper limit of values accurately estimated by LDNe (>10 000, Macbeth *et al.* 2013). Removal of migrants did not considerably affect $\hat{N}_e$ in other lakes. Generally, downstream lakes (Cabot and Esker-WP152) had $\hat{N}_e$s that were approximately an order of magnitude greater than those of headwater lakes (Lake 1, Genetics H and Slushy). Strange had intermediate $\hat{N}_e$s.

T-Bone's negative $\hat{N}_{e(LDNe)}$ was excluded when the *meta*-$\hat{N}_{e(T+H)}$ was calculated with the inclusion of migrants. Migration rates calculated in BAYESASS between lakes without T-Bone did not differ from those calculated for the whole system. *Meta*-$\hat{N}_{e(T+H)}$ with and without migrants (5018.0, 2557.5, respectively) were lower than the sum of the lake $\hat{N}_{e(LDNe)}$ (5797.4, 20534.7, respectively).

$\hat{N}_{e(LDNe)}$ (calculated with migrants) correlated with lake area ($R^2 = 0.920$, $P \leq 0.003$), but only after the negative $\hat{N}_{e(LDNe)}$ calculated for T-Bone was removed. However, neither $\hat{N}_{e(LDNe)}$ (calculated without migrants) nor A_R correlated with lake area ($P \geq 0.471$, $P \geq 0.054$, respectively). There were no significant correlations between elevations and A_R, or between elevations and $\hat{N}_{e(LDNe)}$ with or without migrants (all $P > 0.05$).

Positive $\hat{N}_b$ were estimated for the headwater lakes Genetics H, Slushy and Strange using the fewest number of cohorts (one to three) that resulted in a positive $\hat{N}_b$ (Table 2). The cohorts used to determine $\hat{N}_b$ for Genetics H and Slushy did not contain any migrants. A positive value of $\hat{N}_b$ for Strange only occurred by including migrants. $\hat{N}_{e(adj2)}$ derived from these $\hat{N}_b$ were equal to or greater than each lake's $\hat{N}_{e(LDNe)}$ estimated from a mixture of cohorts.

The best MIGRATE-N model (highest Bezier log marginal likelihood score) allowed for reduced migration within the system in accordance with colonization of the Kogaluk from the west (-6.5×10^7 vs. -8.0×10^7 for the full model).

Median Θ values did not vary greatly between lakes (0.97–1.50) but were notably smaller in headwaters (Lake 1, Genetics H, Slushy and Strange) than downstream lakes (Esker, WP152, and Cabot) with the exception of T-Bone, which demonstrated a higher median Θ of 1.30 similar to that of downstream lakes (Fig. 5).

Median M values varied between 103.3 and 190. There was evidence of historically high migration between the downstream lakes Esker and WP152 as well as high bidirectional migration between Esker and its headwater Slushy. Alternatively, there was greater downstream historical migration from the headwater Strange into the downstream Esker and WP152 than there was upstream migration from these lakes into Strange. There was also evidence of historic migration between the western headwater lakes, particularly between T-Bone and Slushy, and between T-Bone and Strange; but there was less evidence of migration between Slushy and Strange. Historical migration into Cabot from its headwater, T-Bone, was greater than that from its other headwater lakes (Esker, WP152). These results are largely consistent with the most highly supported DIYABC colonization scenario.

Historical colonization

Given their high migration estimates and because DIYABC assumes no migration between populations after divergence (Cornuet *et al.* 2008), Esker and WP152 were combined for this analysis. The pre-evaluation revealed

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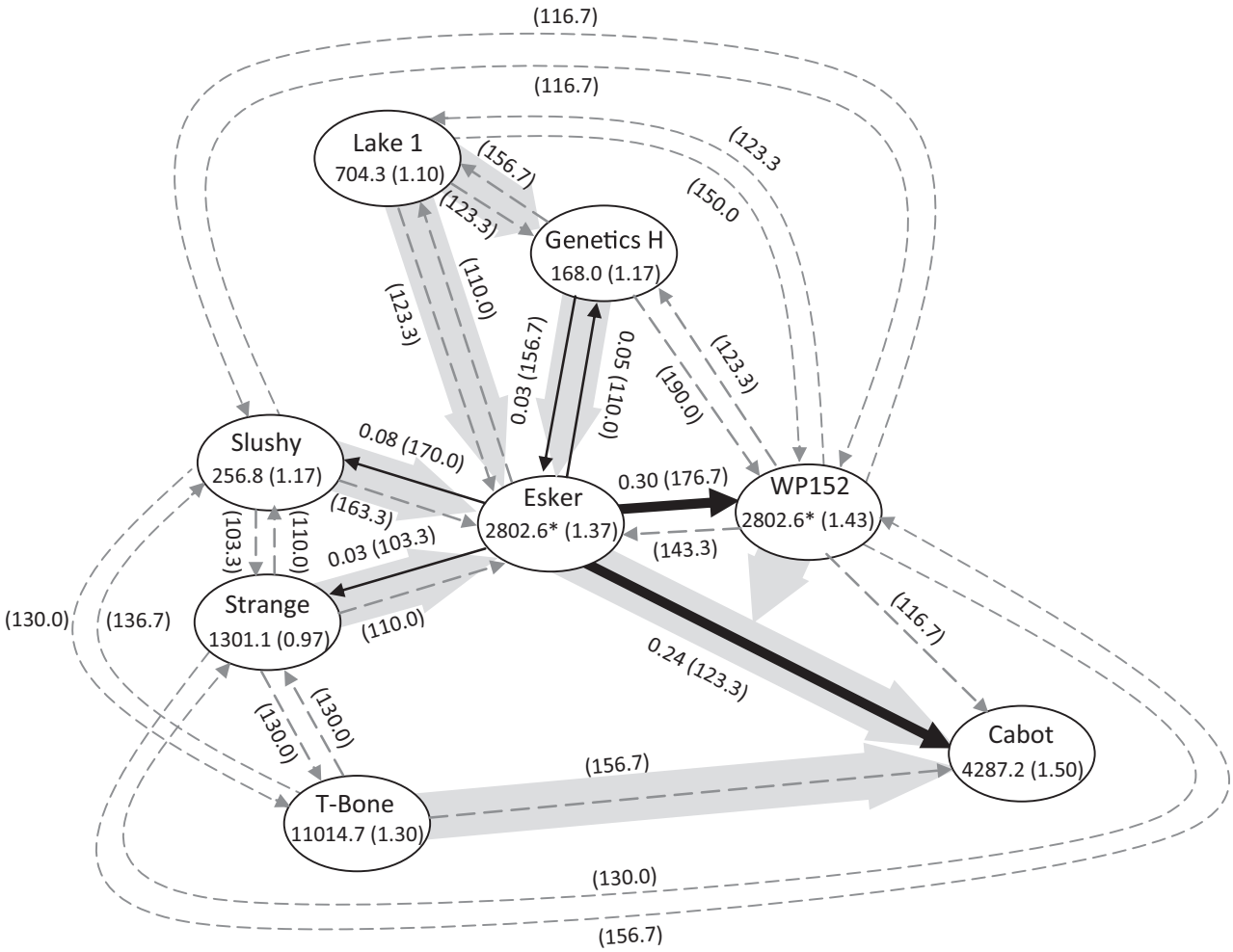

Fig. 5 Contemporary and historical, migration and lake effective sizes within the Kogaluk. Population effective sizes ($\hat{N}_e$) calculated without migrants in LDNe are shown within each lake and historical effective sizes (Θ) calculated using MIGRATE-N are shown in parentheses. Black arrows are significant migration rates (as proportion of individuals per generation) between lakes based on 95% confidence intervals (Rannala 2007) as calculated in BAYESASS (Wilson & Rannala 2003). Thick black arrows indicate migration rates greater than 10% per generation; thin arrows are migration rates less than 10% per generation. Dark, grey, dashed arrows are historical migration rates (*M*) calculated using MIGRATE-N, values are shown in parentheses. Thick light grey arrows are the direction of water flow.

Table 2 The adjusted effective number of breeders ($\hat{N}_{b(adj2)}$), effective population size ($\hat{N}_{e(adj2)}$) of three lakes with positive effective number of breeders ($\hat{N}_b$) calculated according to Waples *et al.* (2014). Values were estimated assuming the age at 50% maturity (α) was 10.7, and adult lifespan (AL) was equal to the maximum age of each particular lake minus 10.7 (α).

Lake	Cohort(s) (by year of birth)	*n*	$\hat{N}_b$	95% CI	$\hat{N}_{b(adj2)}$	$\hat{N}_{e(adj2)}$
Genetics H	2013	78	173.1	113.3–334.3	162.6	269.2
Slushy	2002, 2003	31	815.5	140.7–∞	770.9	1234.0
Strange (with migrants)	2003, 2004, 2005	37	1737.0	184.1 – ∞	1596.9	3000.6

only six summary statistics for which the proportion of values that were lower than that for the observed data set was <0.05 or >0.95. We concluded that the models fit well enough for further analysis. Colonization from the west was the most likely scenario with a posterior probability of 0.7801 (95% CI = (0.5487–1.0000)). In this scenario, the most westerly headwaters (T-Bone, Slushy and Strange) were colonized 615 generations ago (7 380–8 303 years BP assuming a *T* of 12–13.5 which was the range for most lakes except WP152 and Esker) (see

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Table S7, Supporting Information). This date corresponds with significant overflow events from Naskaupi to the Kogaluk between 8400 and 7000 years BP (Jansson & Kleman 2004). Migrants from the western headwaters Strange and Slushy (Strange's admixture rate = 0.208) invaded the downstream Esker-WP152 560 generations ago (6720–7560 years BP). The northern headwaters Lake 1 and Genetics H were colonized by migrants from Esker-WP152 432 generations ago (5184–5832 years BP) and 466 generations ago (5592–6291 years BP), respectively. The most downstream lake, Cabot, was colonized from an equal amount of migrants from the lakes immediately upstream from it (T-Bone and Esker, Esker's admixture rate = 0.449) 210 generations ago (2520–2835 years BP). $\hat{N}_e$ estimated using DIYABC ($\hat{N}_{e(DIYABC)}$) tended to exceed $\hat{N}_{e(LDNe)}$, but relative relationships among lakes remained similar. Type I and type II error remained relatively low for this scenario at 0.168 and 0.055, respectively. Ten of forty-two summary statistics in this scenario were identified as having a proportion of values lower than that of the observed data set <0.05 or >0.95.

Discussion

Life history implications

The maximum age we observed ($\omega = 51$) greatly exceeded that reported by Scott & Crossman (1998) (22–24 years) and that observed in Great Slave Lake ($\omega = 8$) (Harris 1962) and the Willow Creek-Chain Lakes system in southern Alberta ($\omega = 19$) (Walton 1980). The age at 50% maturation, $\alpha = 10.7$, was similar to that of Great Slave Lake (youngest mature fish was 9) (Harris 1962) but greater than that in Lake Superior (youngest mature male and female were 4 and 5 years, respectively) (Bailey 1969) and the Willow Creek-Chain Lakes system (maturation occurred between ages 5 and 6) (Walton 1980). The delayed maturation and longer lifespans observed in this system are consistent with the Kogaluk's high latitude and consequent short seasonal growth period (Blanck & Lamouroux 2007; Munch & Salinas 2009).

Variation in life history traits among lakes could be due to differing predation, competition or habitat quality (Hixon & Jones 2005; Sinsch *et al.* 2007). As a deep fjord, Cabot may provide better thermal habitat for lake trout (*Salvelinus namaycush*) (Christie & Regier 1988), increasing their population size. The resulting increased predation of longnose suckers could have resulted in their observed reduced lifespans in this lake. The greater lifespans, survivorship rates and generation times of the downstream lakes Esker and WP152 may be a result of reduced competition or predation due to their large habitat areas (Hixon & Jones 2005) or alternatively the environmental stability and higher quality habitat associated with downstream lakes (Moore *et al.* 2015).

Overall genetic structure

The genetic distinctiveness of lakes was also observed in lake trout within the Kogaluk, suggesting that streams within this system significantly obstruct gene flow (McCracken *et al.* 2013). This is probably due to the shallowness of these streams, which causes their freezing from late October through April and May (Wheeler 1935). At high temperatures, these streams may also dry up or experience reduced oxygenation (Matthews 1998; Griffiths 2010).

Despite stream transiency and inhospitableness, gene flow was evident among lakes. Contemporary upstream migration from the confluence lake, Esker, is consistent with the migration of spawning adults (McPhail & Lindsey 1970; Ryan 1980; Scott & Crossman 1998). Downstream migration is potentially due to individuals being washed downstream, which particularly affects juvenile longnose suckers (Ryan 1980).

Lake genetic differentiation was clearly influenced by contemporary migration. The genetic distinctiveness of Strange and Lake 1 could be attributed to Strange's shallow outlet and Lake 1's higher altitude and latitude (which could lead to increased freezing of its connecting streams). Higher water volumes in downstream sections of the river probably facilitate the high migration observed between Esker and WP152 (Vannote *et al.* 1980; Ward & Stanford 1995). The high contemporary migration observed from Esker into the downstream fjord lake, Cabot, could be due to washing of juvenile fish down river (Ryan 1980). Alternatively, there may be no migration between these lakes, but their high $\hat{N}_e$s may have prevented detection of their genetic distinction in STRUCTURE, particularly if Cabot was recently colonized from the immediately upstream Esker-WP152 (as supported by DIYABC). DIYABC and MIGRATE-N results suggest Cabot also received historical migration from its other immediate headwater, T-Bone, despite an absence of modern migration detected using BAYESASS. The influence of this historical migration is evident by the position of Cabot in the PCoA between T-Bone, Esker and WP152.

Genetic effects of physical features

Mantel tests and dbRDA both supported the influence of waterway distance on lake genetic differentiation. In contrast to the Mantel tests, there was significant evidence of the influence of waterway distance on genetic

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distance in the dbRDA considering all lakes. This may be a result of the reduced power of Mantel tests relative to dbRDA tests (Legendre *et al.* 2015). However, the significance of the geographic explanatory variables disappeared when elevation and waterfall distances were included as explanatory variables in the dbRDA models.

The marginally insignificant dbRDA and partial Mantel that included waterfalls as the explanatory variable suggest the diminished importance of waterfalls in comparison with waterway distance in this system. Waterfall influence is probably tempered by contemporary and historical downstream migration over waterfalls. However, the lack of upstream migration over waterfalls suggests that the intervening waterfalls probably limit migration between the headwater T-Bone and the northern six lakes. Therefore, the absence of significant effects of waterfalls may also be due to a time lag in the manifestation of their effects on genetic structure (Epps & Keyghobadi 2015). Assuming waterfalls continue to act as barriers to gene flow; T-Bone's genetic divergence from the northern six lakes should increase with time, resulting in a more apparent influence of waterfalls on metapopulation genetic structure.

Neither slope nor elevation was found to influence genetic structure in either dbRDA or Mantel tests. Given the dispersal ability of longnose suckers (McPhail & Lindsey 1970; Ryan 1980; Scott & Crossman 1998), the gradual slopes observed between most lakes probably did not impede their migration. Elevation and slope were similarly inconsequential in shaping lake trout genetic structure within the Kogaluk (McCracken *et al.* 2013), which are also known to be strong swimmers (Scott & Crossman 1998). Slope also did not hinder brook trout (*Salvelinus fontinalis*) within a Connecticut river system (Kanno *et al.* 2011).

Migration–Drift equilibrium

The initial lack of correlation between linearized pairwise $\hat{F}_{ST}$s and geographic distance in the Kogaluk is typical of the case III described by Hutchison & Templeton (1999) expected in highly divided and drift-driven metapopulations. The significant negative correlations between mean linearized pairwise A_R and linearized pairwise $\hat{F}_{ST}$s and residual linearized pairwise $\hat{F}_{ST}$s (after correlation with pairwise distance) further suggest that genetic differentiation is a function of genetic diversity due to allelic loss from drift (Raeymaekers *et al.* 2008).

The IBD pattern (i.e. Hutchison & Templeton's (1999) case I) observed after removing the outlier lakes Strange and Lake 1 indicates that the initial case III relationship was primarily driven by drift in these lakes. This is supported by the insignificant correlation between mean pairwise A_R and $\hat{F}_{ST}$ residuals with the removal of these outlier lakes. The correlation between $\hat{F}_{ST}$ and mean pairwise A_R remains significant after removing Strange and Lake 1 due to the increased drift experienced by these geographically isolated headwaters. However, once geographic distances are taken into account using residual $\hat{F}_{ST}$s the correlation with A_R remains significant only with the inclusion of the outlier lakes, suggesting that these two lakes experience more drift than other headwaters. These results indicate identifying outlier subpopulations is crucial for avoiding the erroneous interpretation of the overall trend in pairwise $\hat{F}_{ST}$s vs. geographic distances.

While lake area correlated significantly with $\hat{N}_e$ (calculated with migrants), it required the omission of one lake from the analysis (T-Bone). In addition, when migrants were excluded, lake area did not correlate with $\hat{N}_e$ nor did it correlate with A_R. The weight of this evidence therefore suggests that habitat area and drift were not the main drivers of genetic diversity in this system.

This varying importance of migration and drift among lakes and the lack of overall IBD suggest that this metapopulation is not in migration–drift equilibrium. This system has many equilibrium-delaying characteristics including its relatively recent colonization (Slatkin 1993), low gene flow between some lakes (Castric & Bernatchez 2003), high lake N_es (Turgeon & Bernatchez 2001), barriers to gene flow (Crispo & Hendry 2005) and the long generation times, slow growth rates, and overlapping generations (Epps & Keyghobadi 2015) exhibited by resident longnose suckers.

Genetic effects of historical colonization

The DIYABC support for colonization from the west is consistent with an invasion of longnose suckers from the palaeolake Naskaupi. This palaeolake has also been proposed as the invasion point of lake chub (*Couesius plumbeus*) into the Kogaluk (Michaud *et al.* 2010). Evidence of historical but not contemporary migration between the western headwater lakes (T-Bone, Strange, and Slushy) also supports colonization of these lakes from Naskaupi. Historical downstream migration from the headwater Strange into the downstream lakes Esker and WP152 was greater than upstream migration, contrary to the upstream (but not downstream) migration observed contemporarily. This is consistent with the proposed colonization of these two lakes from Strange. The headwater Slushy, while also supported as a source of colonists for the downstream Esker and WP152 by DIYABC analysis, demonstrated equal historical migration to and from these lakes. This is possibly due to high

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modern migration from Esker to Slushy but not to Strange.

The lack of migration–drift equilibrium means evidence of colonization could still be present in this metapopulation's contemporary genetic structure (Costello *et al.* 2003). This may explain the high historical Θ and contemporary $\hat{N}_e$ (without migrants) of the most southwest headwater T-Bone which is contrary to its isolated position as a headwater that is further separated by a downstream waterfall. T-Bone's early colonization may have been the source of its genetic diversity as a reduced genetic diversity is expected with successive founding events (Austerlitz *et al.* 1997; Le Corre & Kremer 1998) when a metapopulation is colonized via a stepping-stone model (Kimura & Weiss 1964). Alternatively, it may be due to multiple introductions into T-Bone (Dlugosch & Parker 2008) from prolonged contact with Naskaupi (from 8400 to 7000 years BP) (Jansson & Kleman 2004). After colonization, T-Bone's relatively large lake area could have preserved its genetic diversity despite its modern isolation (Frankham 1996). Bull trout (*Salvelinus confluentus*) populations inhabiting large lakes upstream of a waterfall were similarly found to have greater genetic diversities than bull trout populations inhabiting other relatively small lakes upstream from waterfalls (Costello *et al.* 2003).

Genetic effects of dendritic structure

With the exception of T-Bone, reduced $\hat{N}_e$s calculated using both LDNe and MIGRATE-N were observed in headwaters and a significant negative correlation with A_R and upstream distance was observed within the northern six lakes. This was consistent with the increased isolation expected of headwaters in dendritic metapopulations (Morrissey & de Kerckhove 2009). A decline in A_R from watershed mouth was also observed in dendritic metapopulations of: *Gammarus fossarum* (Alp *et al.* 2012), Trinidadian guppies (*Poecilia reticulata*) (Crispo *et al.* 2006; Barson *et al.* 2009), brook trout (Torterotot *et al.* 2014), three-spined stickleback (*Gasterosteus aculeatus* L.) (Raeymaekers *et al.* 2008) and *Cottus gobio* (Hänfling & Weetman 2006). A reduction in headwater genetic diversity can be caused by dendritic structure (Morrissey & de Kerckhove 2009; Paz-Vinas & Blanchet 2015), sequential upstream colonization (Crispo *et al.* 2006; Caldera & Bolnick 2008) and the larger areas of more downstream habitat patches (Vannote *et al.* 1980; Carrara *et al.* 2014; Paz-Vinas *et al.* 2015). However, in the Kogaluk it is probably predominately due to dendritic structure because the correlation between A_R and lake area was not significant and DIYABC analysis suggested colonization from the headwaters.

The lack of a significant correlation between A_R and upstream distance within the whole system is due to Cabot's low A_R given its downstream position. Cabot's recent colonization could be responsible for a loss in A_R due to the founder effect (Nei *et al.* 1975; Dlugosch & Parker 2008). Alternatively, Paz-Vinas & Blanchet (2015) observed a bell curve in A_R with distance from the outlet in a modelled dendritic metapopulation when the most downstream population was geographically distant (as Cabot is in this system).

Contrary to the predictions of a dendritic system, we did not find a correlation between elevation and $\hat{N}_e$. This may be attributed to the minimal elevation differences among most lakes. Additionally, evidence of a correlation of A_R but not H_o or H_e with upstream distance in the Kogaluk may be due to A_R achieving equilibrium and reflecting recent landscape changes before heterozygosity (Epps & Keyghobadi 2015). The number of alleles but not heterozygosity was similarly found to decrease with distance from the outlet in Trinidadian guppies (Crispo *et al.* 2006). This suggests that the full influence of the dendritic nature of the watershed on this metapopulation's genetic structure has not been realized.

Perhaps as a result of this delayed realization of the effects of the dendritic structure, there was no evidence for an elevated metapopulation genetic diversity in the Kogaluk. The sum of the individual lake $\hat{N}_{e(LDNe)}$s always exceeded the *meta*-$\hat{N}_{e(T+H)}$. Support for this result is somewhat tempered by the large confidence intervals associated with most of our $\hat{N}_{e(LDNe)}$s. However, the observation that $\hat{N}_{e(adj2)}$ was greater than $\hat{N}_{e(LDNe)}$ in those lakes for which a positive $\hat{N}_b$ was estimated suggests that $\hat{N}_{e(LDNe)}$ were conservative estimates of the actual N_e values and that the sum of $\hat{N}_{e(LDNe)}$s may therefore be even greater than estimated. The lack of an overall elevation on genetic diversity could be due to the Kogaluk's spatial simplicity as increased branching is expected to maximize genetic differences between headwaters and confluences (Mari *et al.* 2014; Yeakel *et al.* 2014). Alternatively, upstream migration from the confluence lake, Esker, could have lessened genetic differences among headwaters (Mills & Allendorf 1996). However, upstream-biased migration may also prevent headwater and metapopulation extinction (Ewers & Didham 2006; Barson *et al.* 2009; Campbell Grant 2011). In natural systems, the required degree of headwater isolation to increase metapopulation N_e may be counteracted by an increased risk of headwater extinction which would decrease metapopulation N_e (Ewers & Didham 2006; Campbell Grant 2011). The lingering influence of the historical concentration of genetic diversity in the headwaters (particularly in T-Bone) due to a lack of migration–drift equilibrium has also probably prevented realization of

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the full genetic effects of the dendritic structure of the system. However, migration–drift equilibrium might never be attained given the many factors operating against its achievement in this system. In addition, we suggest migration–drift equilibrium is probably to elude many other high-latitude metapopulations due to their recent colonizations (Slatkin 1993) after the last glacial maximum and the long generation times and slow growth associated with populations inhabiting high-latitude environments (Blanck & Lamouroux 2007; Munch & Salinas 2009).

Given these many factors in nature that can conflict with the effects of dendritic structure, we suggest few, if any, dendritic metapopulations will achieve an elevated genetic diversity. The dendritic metapopulation structure may be similar to Wright's island model; an idealized model that does not appear in nature (Whitlock & Barton 1997; Waples 2010). Our results therefore support the assertion that fragmentation generally decreases metapopulation $\hat{N}_e$ (Whitlock & Barton 1997; Waples 2010; Palstra & Ruzzante 2011; Gomez-Uchida *et al.* 2013).

Conclusion

Our results suggest that the conditions under which the fragmentation of dendritic metapopulations is expected to confer added genetic diversity and resistance to extinction are limited given the complex interaction of factors affecting these metapopulations in nature. This will be particularly true in those dendritic metapopulations lacking migration–drift equilibrium. Such metapopulations may be more vulnerable than their neutral genetic structures suggest (Lippe *et al.* 2006) whether the fragmentary influences of some contemporary factors (e.g. waterfalls) have not yet been fully genetically realized due to a time lag (Epps & Keyghobadi 2015). This has important implications for future modelling studies and the conservation of dendritic metapopulations. Despite a growing understanding for the need to develop a theoretical framework for dendritic metapopulations (Campbell Grant *et al.* 2007; Labonne *et al.* 2008; Campbell Grant 2011; Baguette *et al.* 2013), present models make oversimplifying assumptions (Perkin & Gido 2012) that do not consider environmental, biological and historical factors. Our results confirm the necessity of considering the dynamic effects of these factors on both the whole system and individual subpopulations for a comprehensive understanding of a metapopulation.

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Data accessibility

Genetic, geographic and elevation pairwise distances among lakes, as well as microsatellite scores can be found in DRYAD digital depository doi:10.5061/dryad.5n7f6.

Supporting information

Additional supporting information may be found in the online version of this article.

Appendix S1. Life history analyses conducted using age correction.

Table S1.1. Maximum age (ω), adult lifespan (AL), Robson-Chapman survival estimate ($\hat{S}$), and generation time (T) of longnose suckers (*Catostomus catostomus*) in eight lakes within the Kogaluk River.

Table S1.2. Kolmogorov-Smirnov test results comparing corrected age class distributions between years in each lake.

Table S1.3. The adjusted effective number of breeders $\hat{N}_{b(adj2)}$ and adjusted effective population size $\hat{N}_{e(adj2)}$ of three lakes with positive effective number of breeders ($\hat{N}_b$) calculated according to Waples *et al.* (2014).

Fig. S1.1. Age at 50% maturity (α) for longnose suckers (*Catostomus catostomus*) estimated using n = 1072 samples from the Kogaluk using a binomial logistic regression.

Appendix S2. Methods analyses details.

Appendix S3. Genetic quality control details.

Table S1. Primers used to genotype longnose sucker samples from the Kogaluk River in Labrador, Canada.

Table S2. PCR reaction reagents for amplification of longnose sucker genetic samples.

Table S3. Thermocycling conditions for amplification of longnose sucker genetic samples.

Table S4. Kolmogorov-Smirnov test results comparing age class distributions between years in each lake.

Table S5. Results of Mantel tests correlating pairwise waterway distances in km (D), elevation differences in m (E), slope differences in m/km (S), and number of waterfalls (WF), with pairwise linearized $\hat{F}_{ST}$s (G) for longnose sucker (*Catostomus catostomus*) samples from eight lakes within the Kogaluk River.

Table S6. Results of distance-based redundancy analyses conducted to compare G genetic distance principal coordinates based on pairwise $\hat{F}_{ST}$s between lakes of longnose suckers (*Catostomus catostomus*) with environmental variables: waterway distances between lakes (D), elevation difference between lakes (E), waterfall distance between lakes (WF).

Table S7. Prior ranges and mean posterior values for the estimated parameters of the best colonization scenario ("colonization from the west") attributed to a metapopulation of longnose suckers (*Catostomus catostomus*) in the Kogaluk River.

Fig. S1. The age at 50% maturity (α) for longnose suckers (*Catostomus catostomus*) from the Kogaluk River estimated using a binomial logistic regression of n = 1072 samples of age vs. maturity.

Fig. S2. Correlation between (a) H_E, (b) H_O, and (c) A_R and distance from the most downstream lake (Cabot).

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